

Project Title:

Calculator using Python

Introduction:

A calculator is a simple device that helps in doing arithmetic calculations like addition, subtraction, etc. This project is about making a basic calculator program in Python. The program takes input from the user and gives the correct result. Python is chosen because it is easy to learn, simple to read, and very useful for programming beginners.

Objectives:

1. To create a calculator that works in the command line.
2. To perform basic operations like addition, subtraction, multiplication, division, modulus, floor division, and power.
3. To handle mistakes such as division by zero.
4. To make the program simple and user-friendly.

Tools used:

Language: Python

Platform: Visual Studio Code (VS Code)

Operating System: Windows 10.

Methodology:

The calculator is made using functions in Python. Each operation has its own function:

- a. `add(x,y)` - Adds two numbers.
- b. `subtract(x,y)` - Subtracts the second number from the first.

- c. multiply(x,y) - Multiplies two numbers.
- d. divide(x,y) - Divide first number by the second (and shows error if the second number is zero.)

The program uses a menu system with a loop, so the user can do many calculations until they choose to exit.

Results and Discussions:

- 1. The calculator works correctly for all 7 basic operations.
- 2. Division by zero is handled properly with an error message.
- 3. The program runs continuously until the user exits.

Conclusion :

This project shows how Python functions, loops, and conditions can be used to make a simple calculator. It meets the project goals and is easy for users to operate. This project can also be used as a base for creating bigger and more advanced applications.

CONCLUSION (FROM MY POV)

From the developer's or my perspective, this project was a good opportunity to practice and understand the basic concepts of Python programming. While creating this calculator, I applied functions, loops, and conditional statements to solve real problems.

The project was also helpful in learning how to handle errors and exceptions, such as division by zero, without stopping the program. Writing the code in a structured way with separate functions for each operation made the program easy to read, maintain, and extend.

This project proved that even a simple idea like a calculator can help improve programming skills. It gave me confidence to build more complex projects in the future, such as a scientific calculator or a GUI-based calculator using Tkinter.